

The percentage of volume *POV* variable measures the amount of market volume the order participated with over the trading period. For example, if a trader executes 20,000 shares of stock over a period where 100,000 shares traded in the market (including the order) the *POV* rate is $20,000/100,000 = 20\%$. *POV* is a very intuitive measure. For example, $POV = 25\%$ means the order participated with 25% of market volume, and $POV = 100\%$ means that the trader accounted for all the market volume during this period. *POV* is the preferred trading strategy metric when monitoring current and historical activity.

The disadvantage of the *POV* strategy is that it contains a decision variable in the denominator, which creates an additional layer of mathematical complexity during trade strategy optimization and increases the solution time.

Trading Rate

$$\alpha = \frac{X}{V_t} \quad \alpha \geq 0 \quad (7.5)$$

The trading rate variable α is the ratio of the shares traded X to the market volume V_t during the trading period, excluding its own traded shares. For example, if a trader executed 20,000 shares in the market over a period of time when 100,000 shares traded in the market; 20,000 shares were from the investor's order and 80,000 shares from other participants, so the trading rate is $\alpha = 20,000/80,000 = 25\%$. If a trader executed 20,000 shares in the market over a period of time when 30,000 shares traded in the market; 20,000 shares were from the investor's order and 10,000 shares from other participants, so the trading rate is $\alpha = 20,000/10,000 = 200\%$.

Trading rate, unfortunately, is not as intuitive as *POV* rate. A trade rate of $\alpha = 100\%$ does not mean that the trader participated with 100% of market volume but rather the investor participated with 50% of market volume. The advantage of the trade rate is that it does not have a decision variable in the denominator so trading solution calculations are less complex and optimization processing time is much quicker. Trading rate is the preferred metric when forecasting costs and developing single stock optimal trading strategies.

Trade Schedule

The trade schedule x_k strategy defines exactly how many shares to transact in a given trading period. For example, the trade schedule for an order executed over n -period is

$$x_1, x_2, x_3, \dots, x_n \quad (7.6)$$